

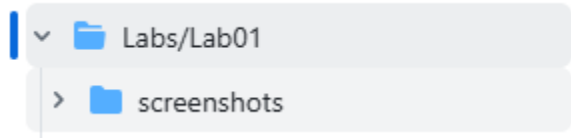
Fatima Jinnah Women University



**Department of Computer Science
DevOps Lab # 1**

**Name: Hina Ali
Registration no: 2023-BCS-040
Course code: CS-649
Section: BCS VII(B)
GitHub Username: Hina-Ali32**

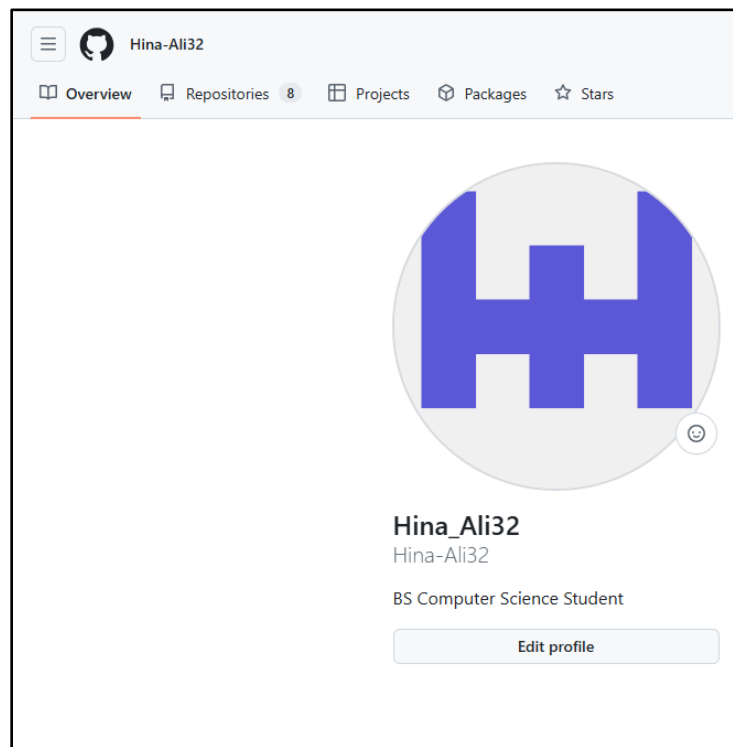
Getting Started



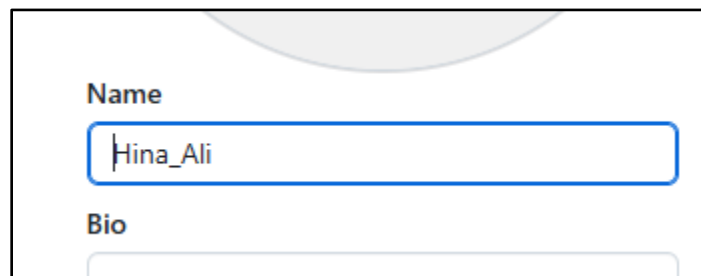
The directory Labs/Lab01/screenshots was created inside the DevOps repository.

Task 1: Create and Configure Your GitHub Account

A GitHub account was created using a personal email and a professional username related to the actual name. The complete actual name was then added in Settings > Public profile.



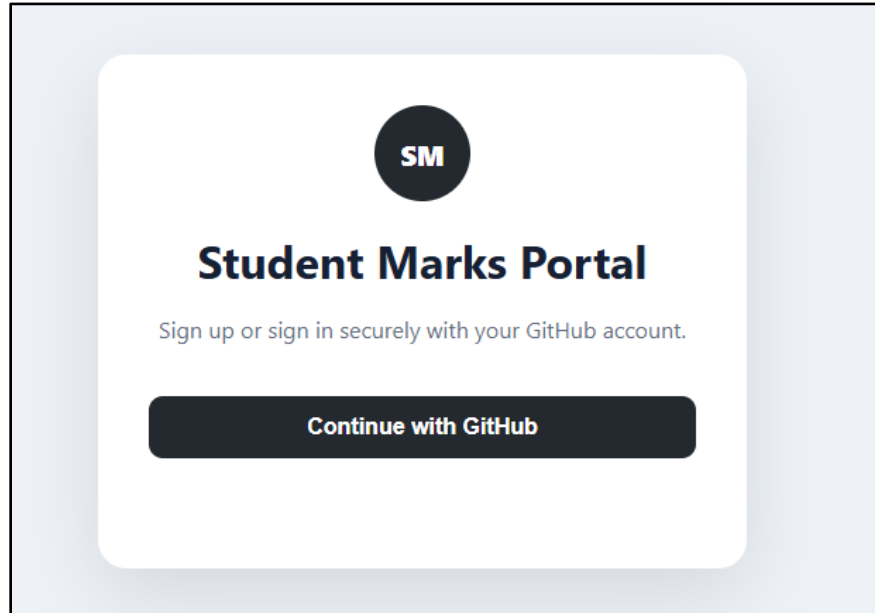
Screenshot showing the GitHub profile with my name and username.



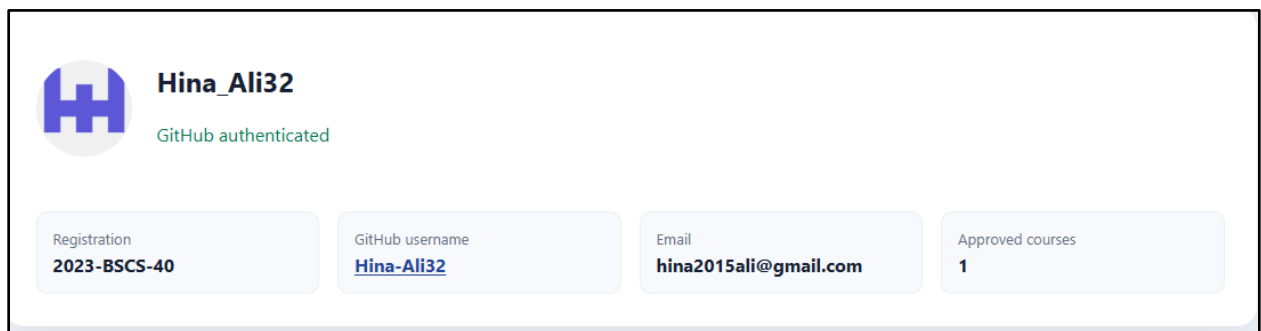
Screenshot showing the complete actual name entered in the Name field.

Task 2: Register in the Student Marks Portal

The Student Marks Portal was opened and sign-in was completed with Continue with GitHub using the account from Task 1. The registration number, course code and section were entered, and the enrollment was submitted for approval.



Screenshot showing the Student Marks Portal login page.



Shows the GitHub profile information inside the portal.

STUDENT MARKS PORTAL

Academic Record

Sign out



Hina_Ali32

GitHub authenticated

Registration

2023-BSCS-40

GitHub username

[Hina-Ali32](#)

Email

hina2015ali@gmail.com

Approved courses

1

+ Enroll in another course

APPROVED COURSES

Course Marks

Not published

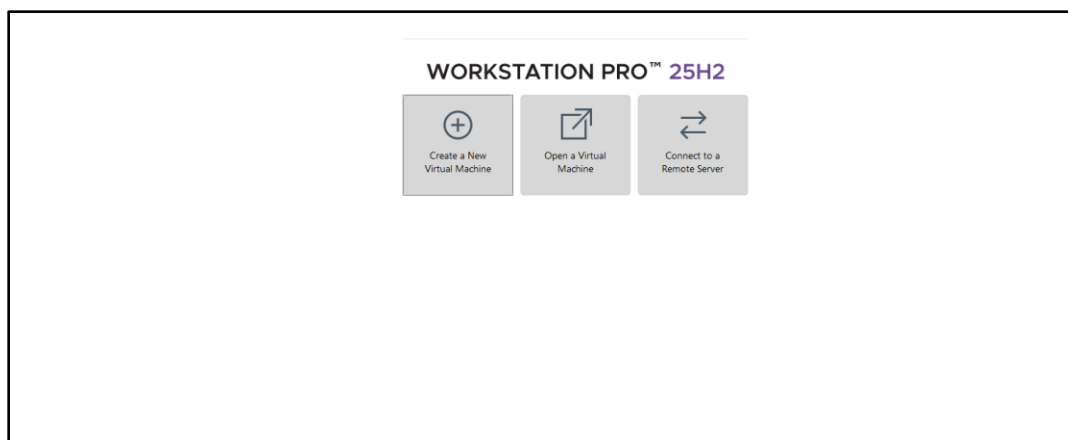
Cloud DevOps (CS-649) Section A

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Marks have not been published.			
Total	—	—	—

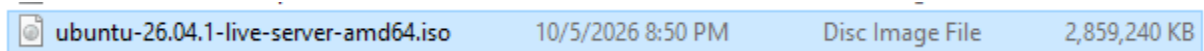
Shows the enrollment submitted and pending approval.

Task 3: Download VMware Workstation Pro and Ubuntu Server

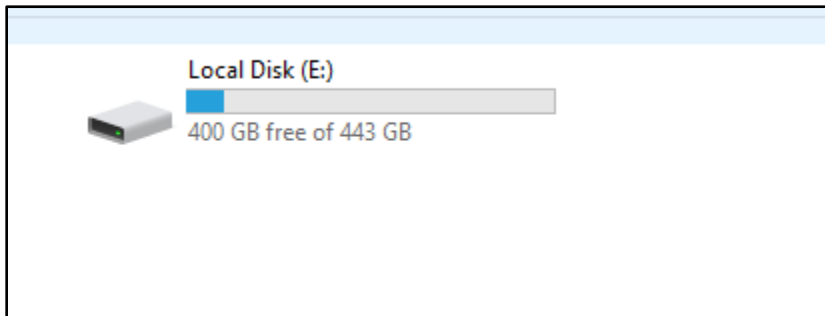
VMware Workstation Pro was already installed, so it was opened to verify that it works. The latest Ubuntu Server LTS ISO was downloaded, and more than 20 GB of free storage was confirmed.



Shows VMware Workstation Pro running.



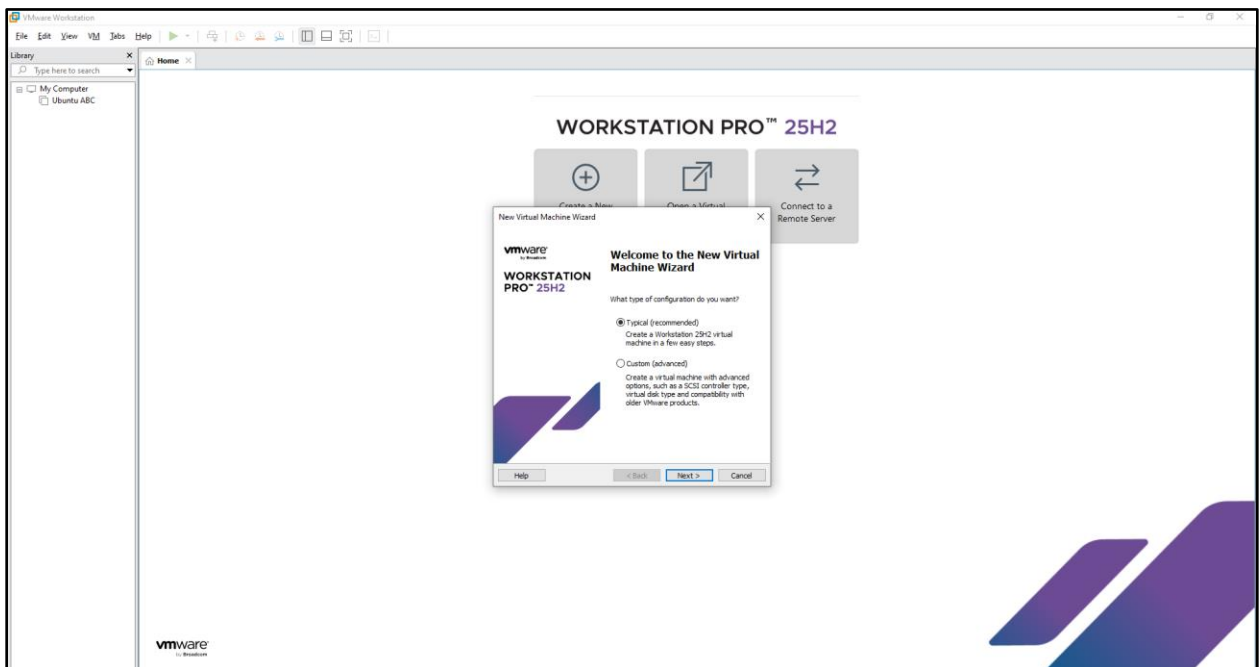
Shows the downloaded Ubuntu Server ISO filename.



Shows the available storage.

Task 4: Create the Ubuntu Server Virtual Machine

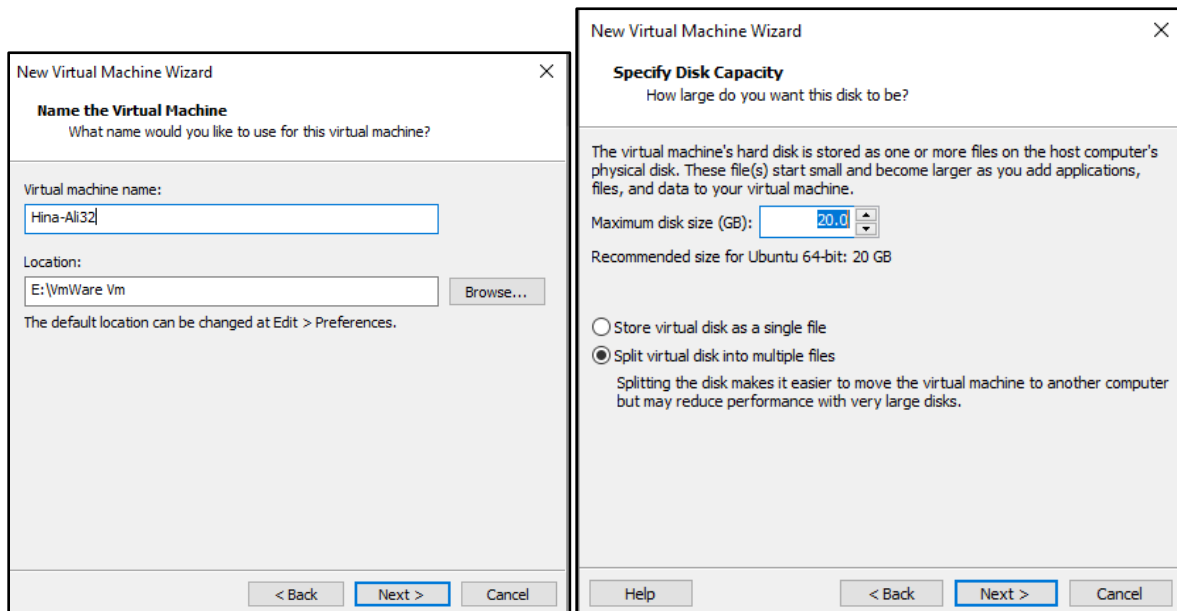
A new virtual machine was created in VMware Workstation Pro using the Typical configuration and the Ubuntu Server ISO, with at least 20 GB of disk space. The virtual machine was then powered on until the Ubuntu Server installer started.

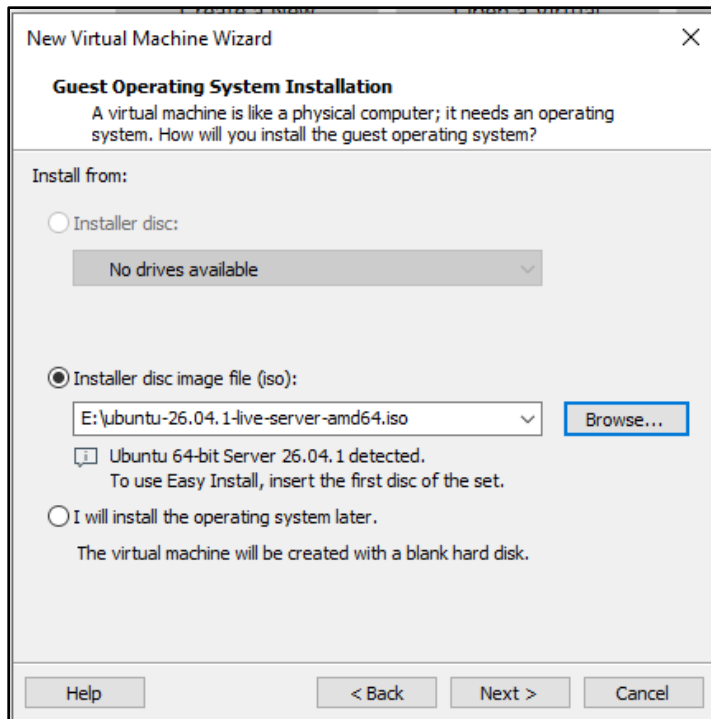


Shows the New Virtual Machine Wizard.

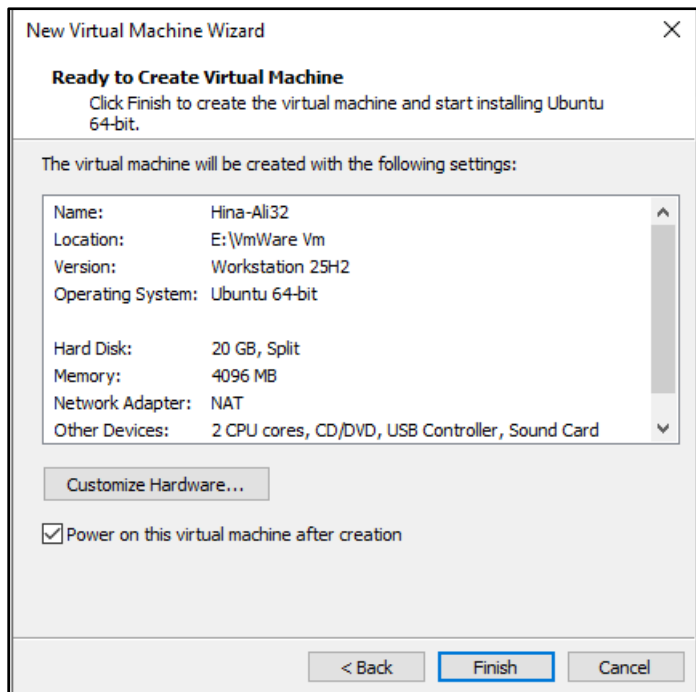


Shows the Typical configuration selected.

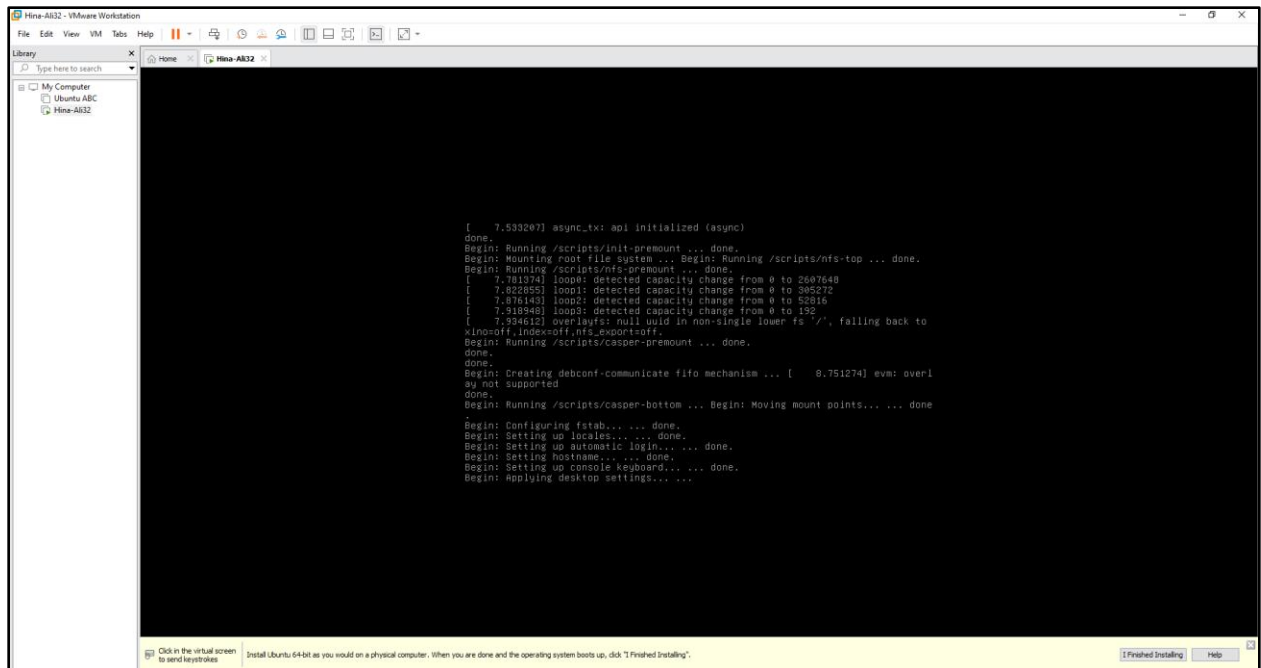




Shows the Ubuntu Server ISO selected



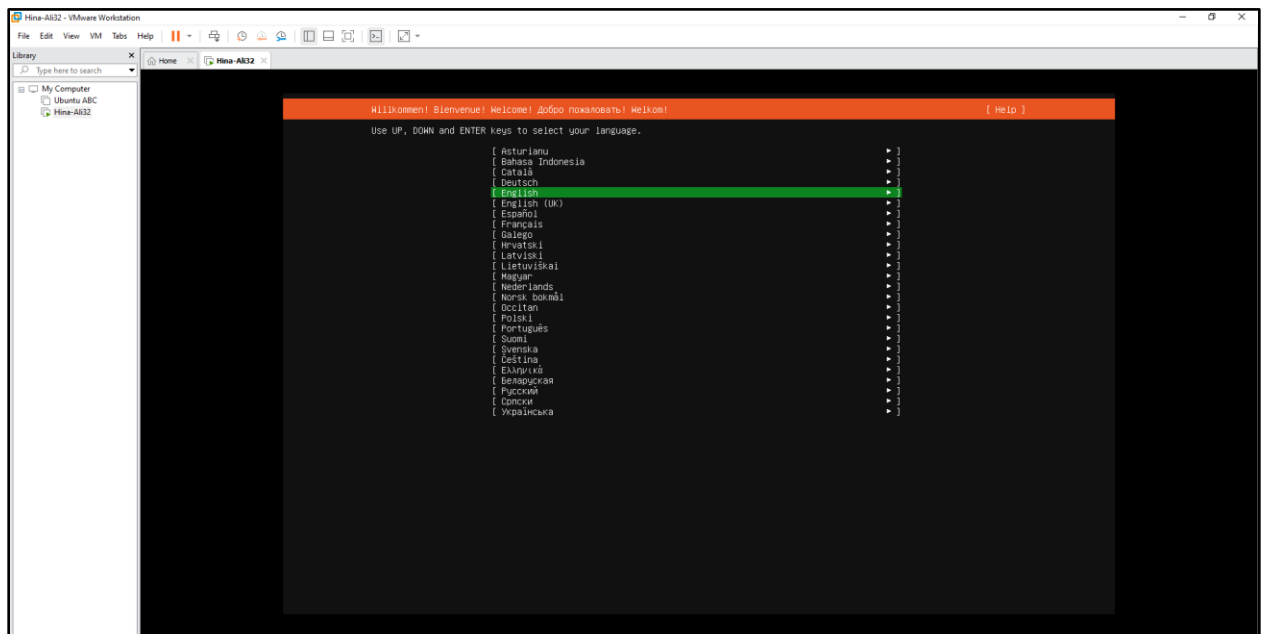
Shows the final virtual machine configuration.

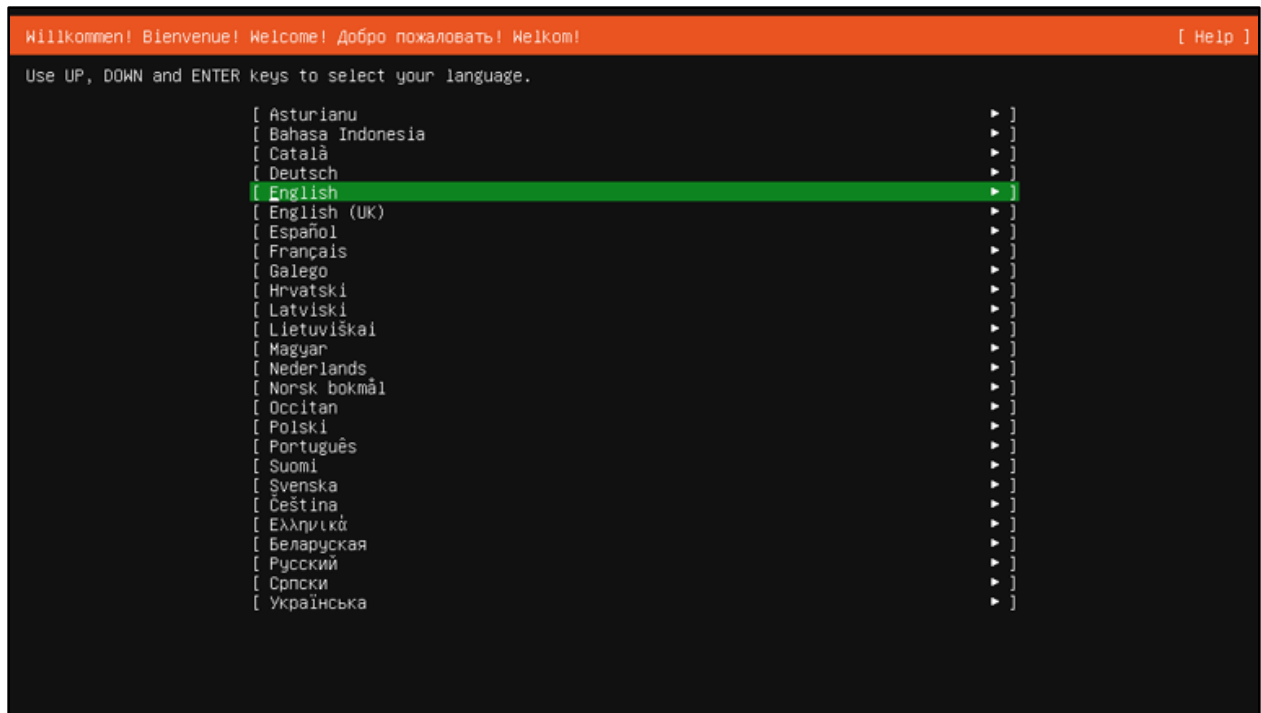


Shows the Ubuntu Server installer boot screen.

Task 5: Choose the Installation Language

The installation language was selected with the arrow keys, and Enter was pressed to continue.

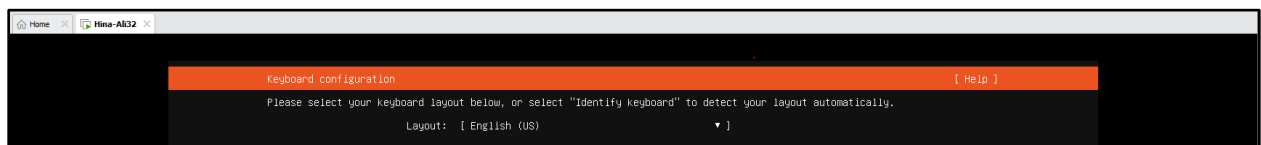




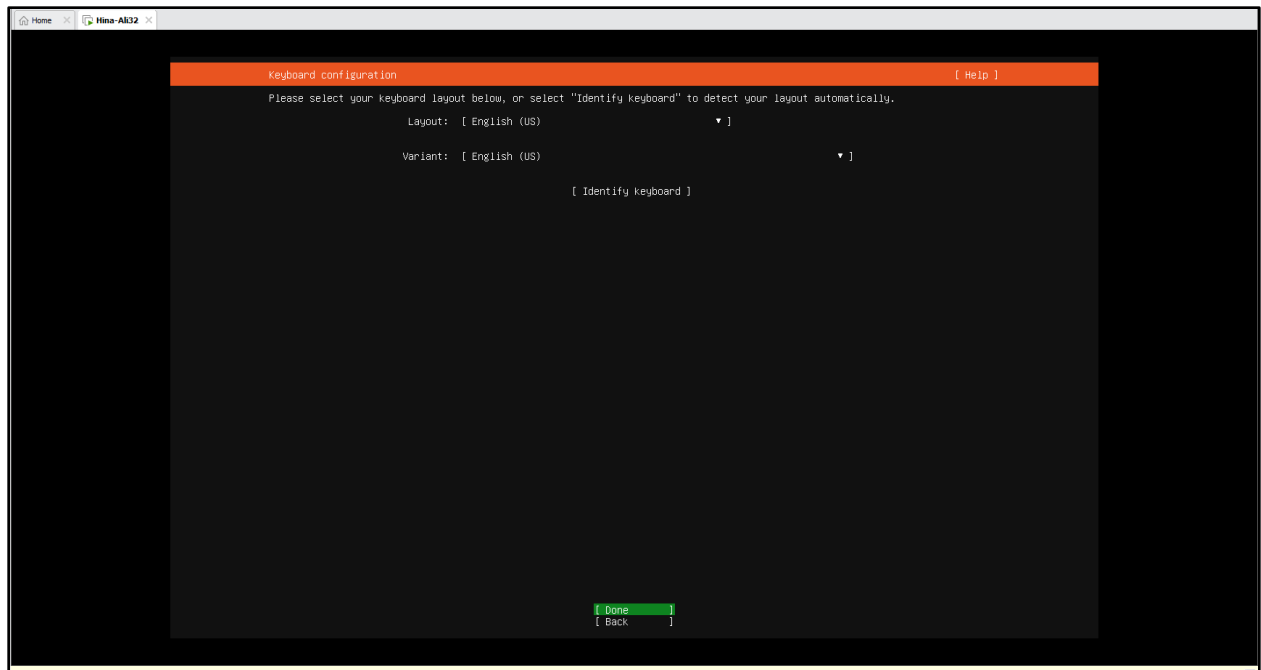
Shows the selected installation language.

Task 6: Configure the Keyboard Layout

The keyboard layout and variant matching the keyboard were selected, followed by Done.



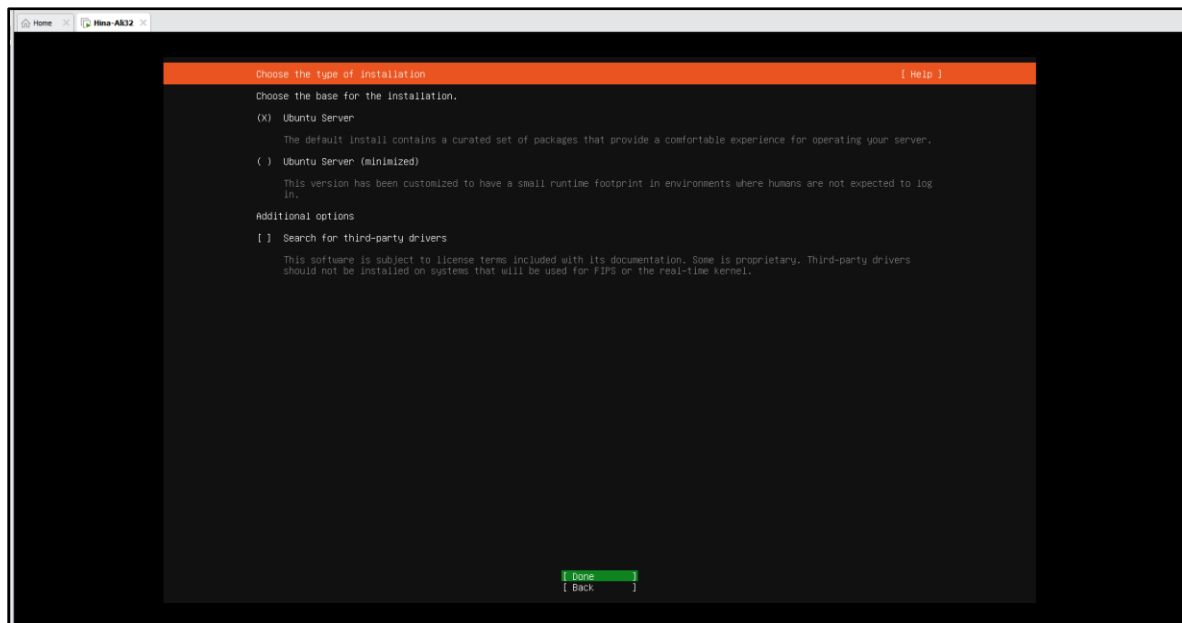
Shows the selected keyboard layout.



Shows the selected keyboard variant

Task 7: Choose the Ubuntu Server Installation Type

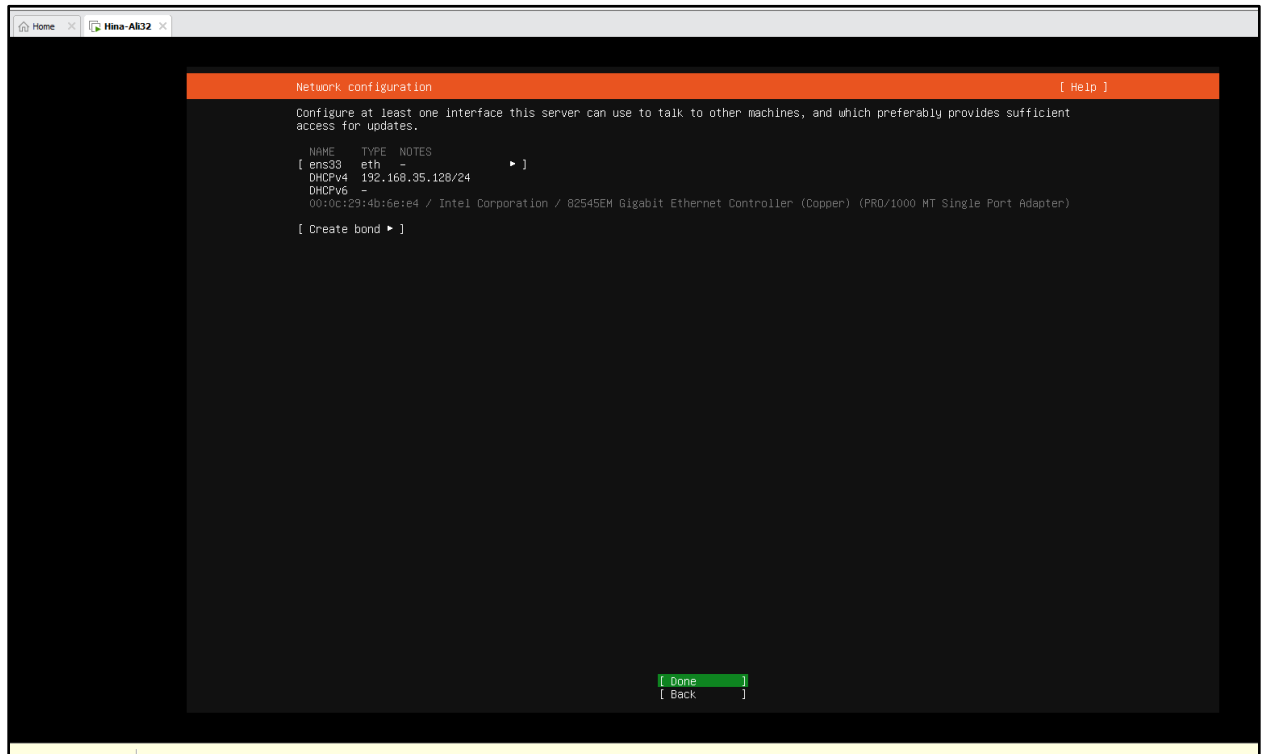
The normal Ubuntu Server installation was selected. MAAS was not selected.



Shows the selected installation type.

Task 8: Configure the Network

The network interface detected by the installer was reviewed, and the automatic IP address from DHCP was confirmed.



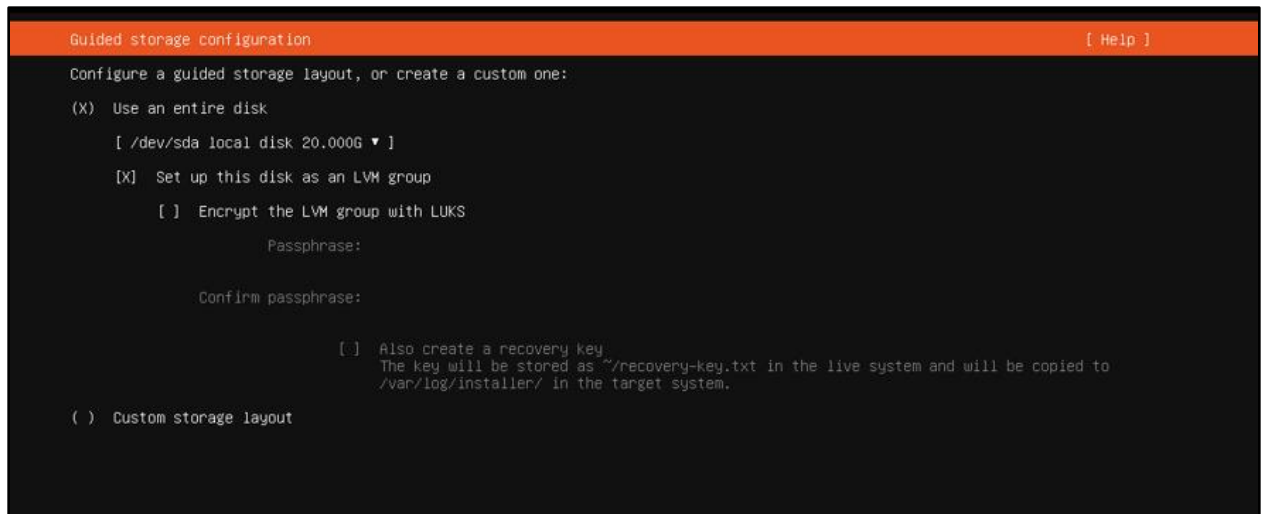
Shows the detected network interface.



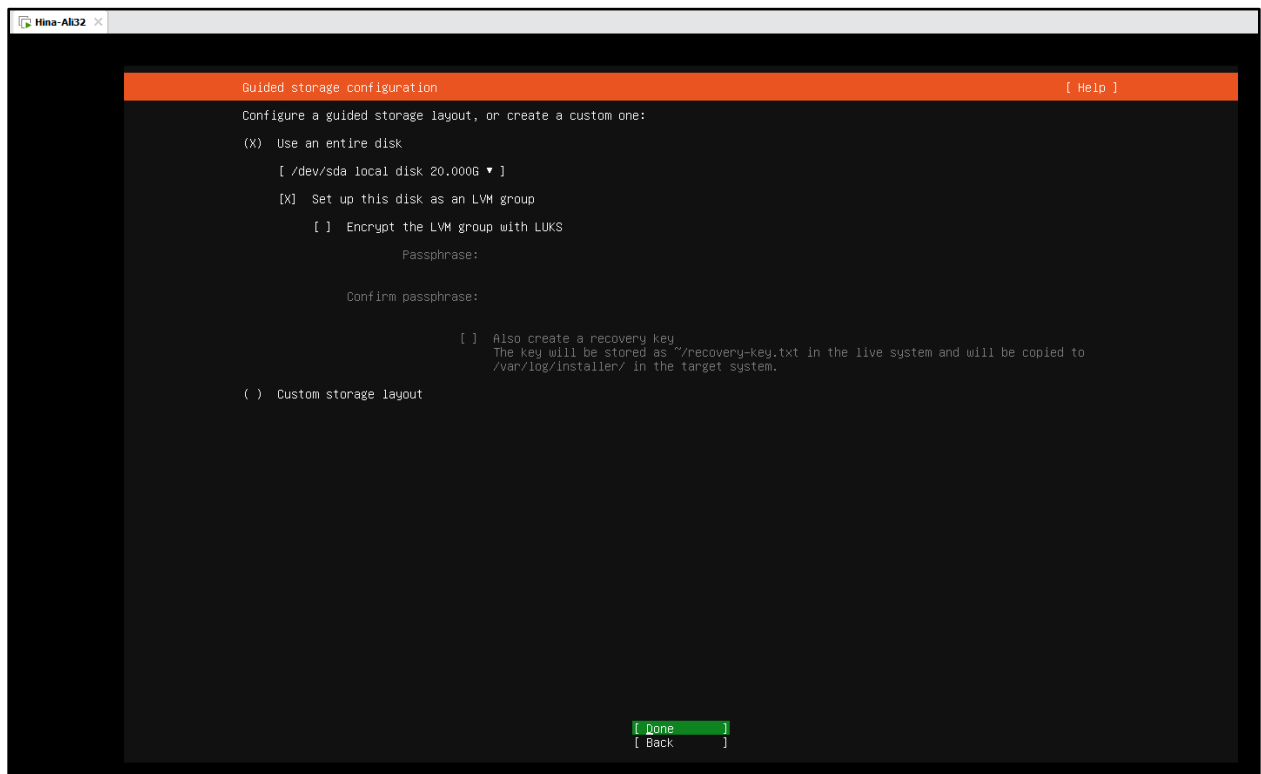
Shows the IP address assigned by DHCP.

Task 9: Configure Storage

The guided storage option was selected to use the virtual disk, and the storage configuration was reviewed.



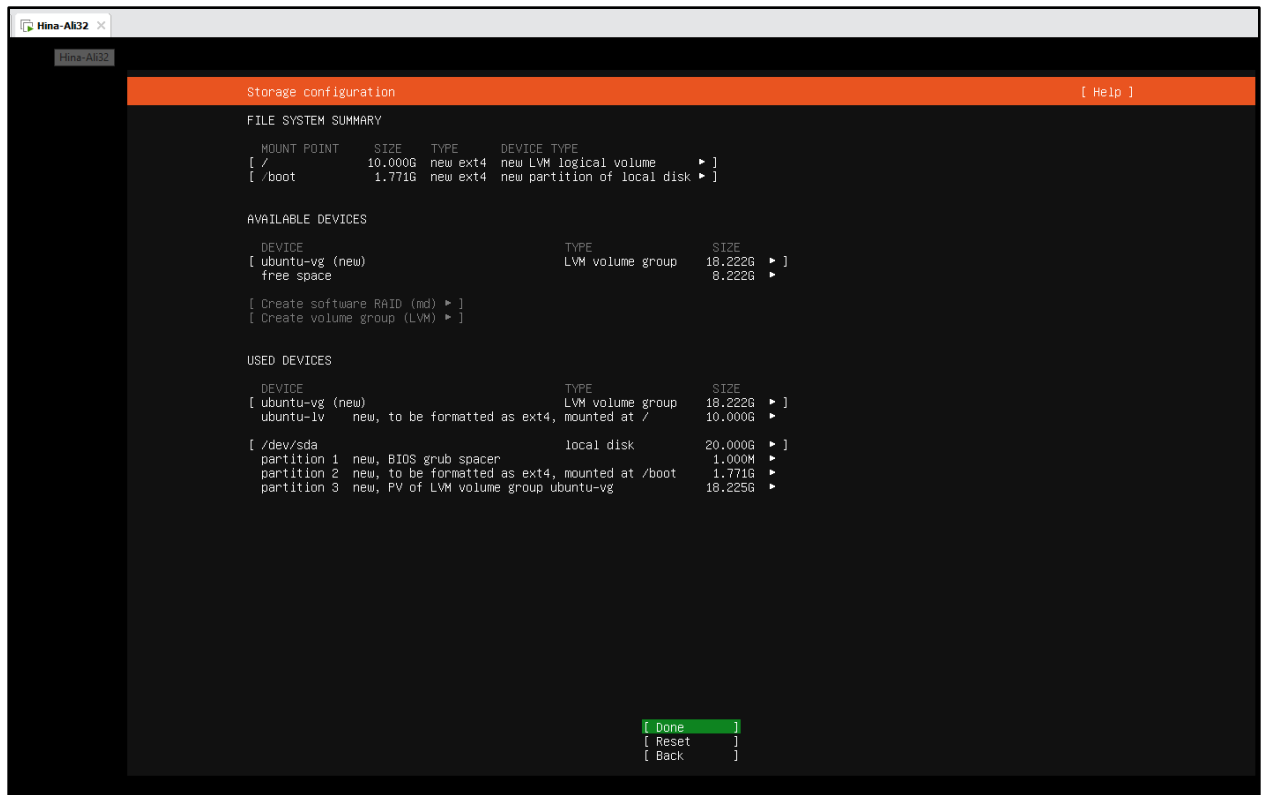
Shows the selected guided storage option.



Shows the storage configuration.

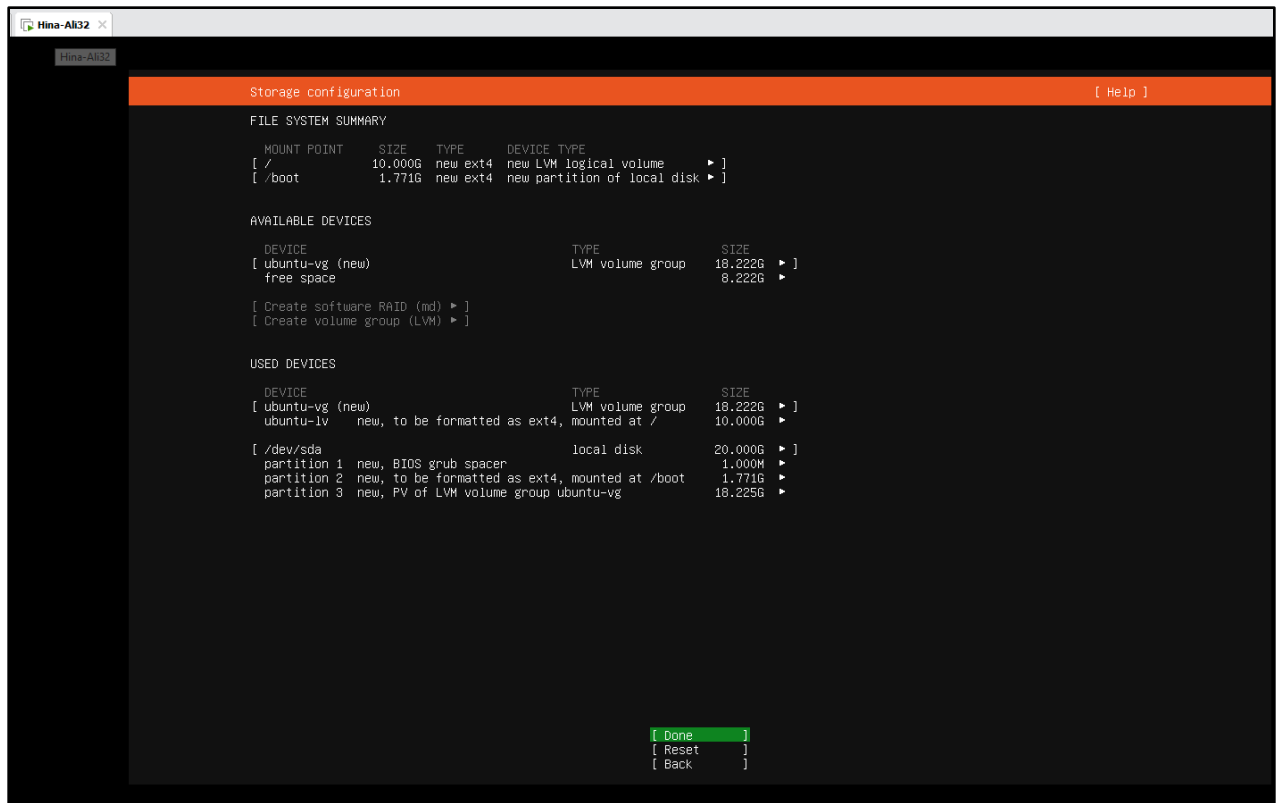
Task 10: Select the Installation Disk

The virtual disk created for this virtual machine was selected, and its name and size were checked.



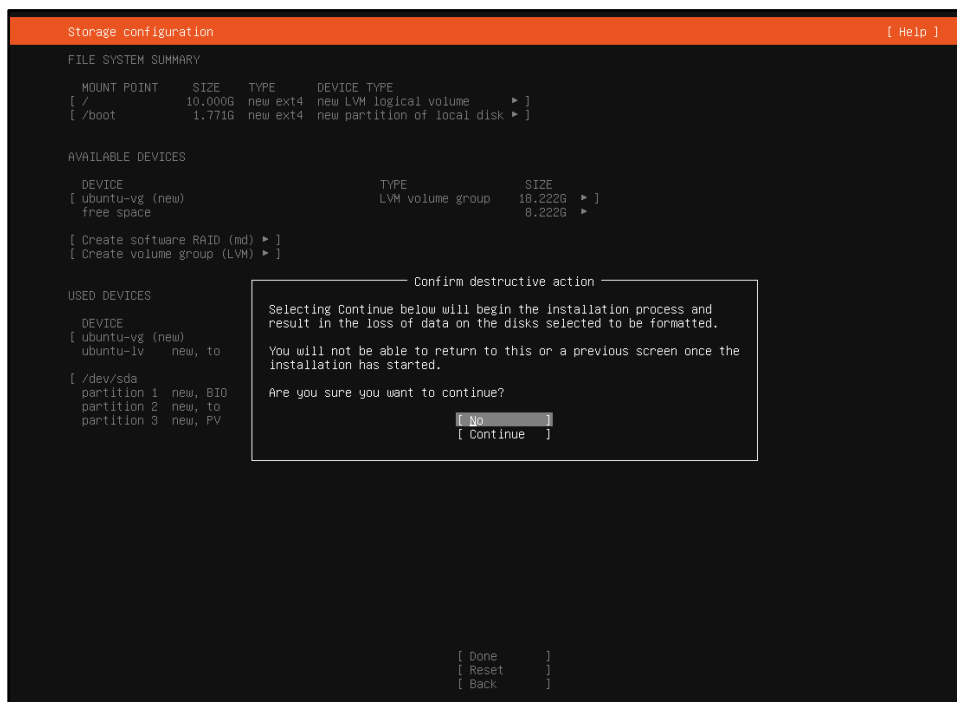
Task 11: Review the Partitions

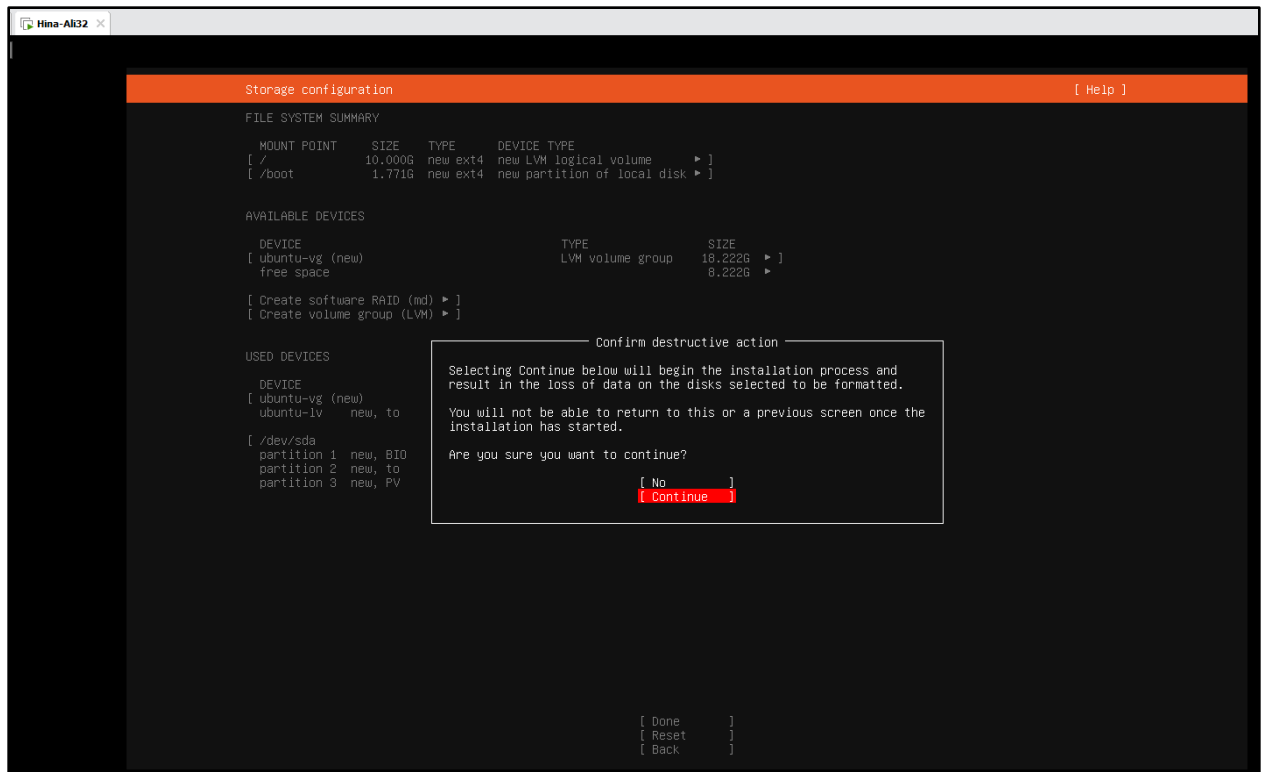
The partition layout proposed by the installer was reviewed and confirmed to be on the virtual disk before selecting Done.



Task 12: Confirm the Storage Changes

The warning about the storage changes was read, and the changes were confirmed after checking that the correct virtual disk was selected.

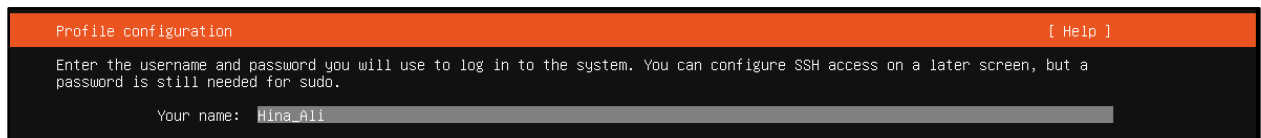




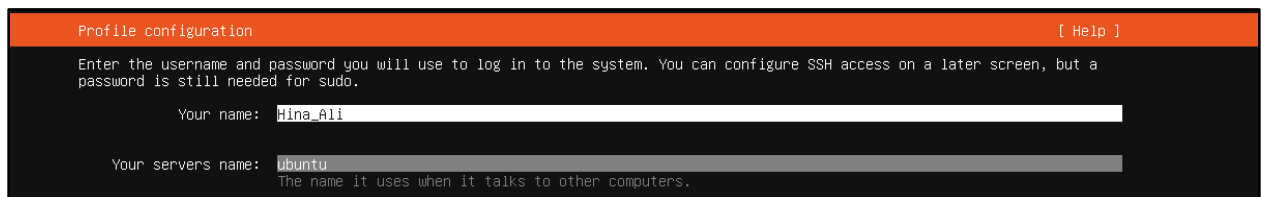
Shows the storage changes confirmed.

Task 13: Configure the Ubuntu Profile

The profile was completed with the complete actual name, the server name ubuntu, the exact GitHub username as the username, and a secure password.



Shows the actual name entered.



Shows the server name ubuntu.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Shows the GitHub username as the Ubuntu username.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

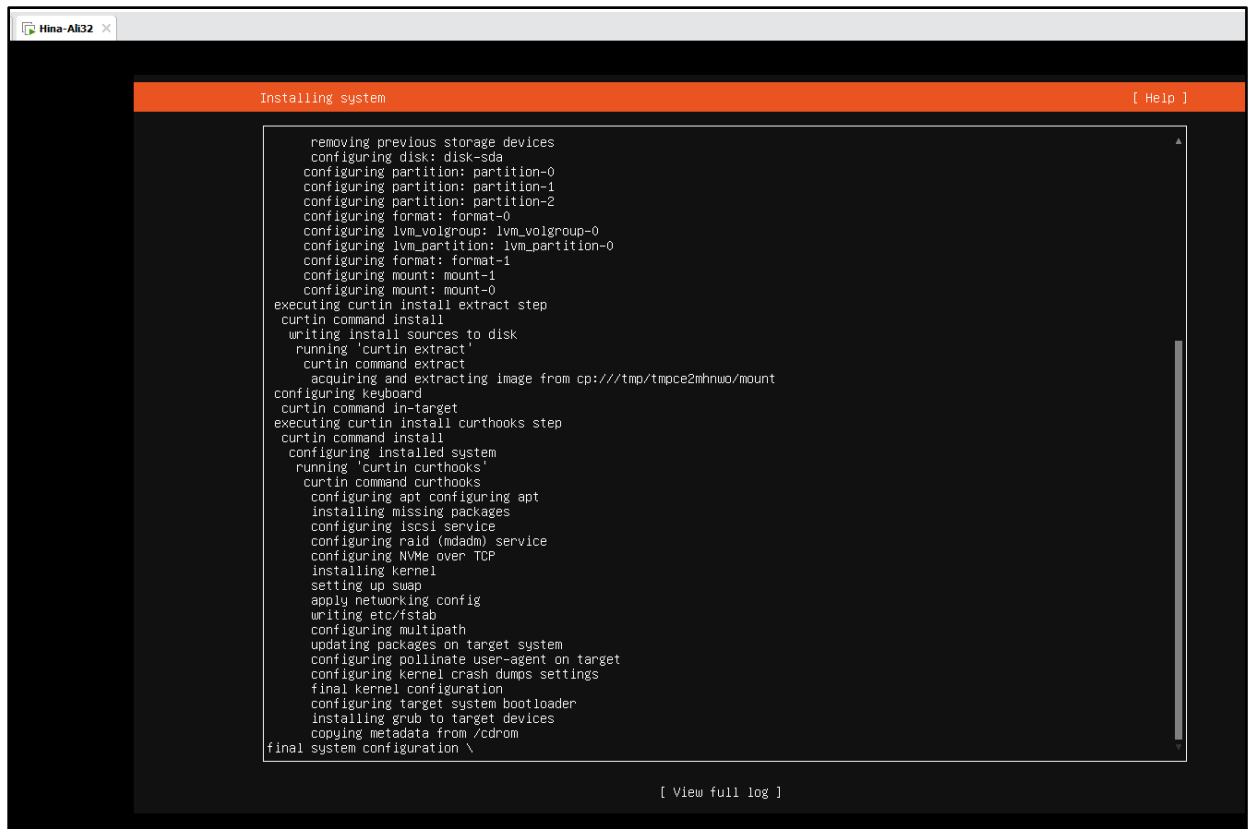
Choose a password:

Confirm your password:

Shows the completed profile page.

Task 14: Complete the Software Installation

Ubuntu Server installed the required software while the virtual machine stayed powered on.



```
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpce2mhnwo/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration \
```

[View full log]

Task 15: Reboot and verify the Ubuntu Identity

After the installation finished, Reboot Now was selected and the installation media was disconnected. Login was completed with the configured username and password, and the prompt showed username@ubuntu:~\$.

```
Installation complete!

writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpce2mhnwo/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
curtin command in-target
curtin command in-target
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
  configuring cloud-init
  restoring apt configuration
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

[View full log]
[Reboot Now]

Shows the installation complete screen.

```
ubuntu login:
```

Shows the Ubuntu login screen.

```
ubuntu login: hina-ali32
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Mon Oct  5 05:51:25 PM UTC 2026

System load:  0.08               Processes:            216
Usage of /:   29.1% of 9.75GB    Users logged in:     0
Memory usage: 11%               IPv4 address for ens33: 192.168.35.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Shows the successful terminal login.

```
hina-ali32@ubuntu:~$ _
```

Shows the prompt username@ubuntu:~\$ verifying the identity.

Task 16: Find the Ubuntu Server IP Address

The command `ip addr` was run, and the IPv4 address of the active network interface was found. The loopback address 127.0.0.1 was ignored.

```
hina-ali32@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:4b:6e:e4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c294b6ee4
    inet 192.168.35.129/24 metric 100 brd 192.168.35.255 scope global dynamic ens33
        valid_lft 1254sec preferred_lft 1254sec
    inet6 fe80::20c:29ff:fe4b:6ee4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hina-ali32@ubuntu:~$
```

Shows the ip addr command and its output.

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:4b:6e:e4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c294b6ee4
    inet 192.168.35.129/24 metric 100 brd 192.168.35.255 scope global dynamic ens33
        valid_lft 1254sec preferred_lft 1254sec
    inet6 fe80::20c:29ff:fe4b:6ee4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hina-ali32@ubuntu:~$
```

Shows the IPv4 address of the active network interface.

Task 17: Connect to Ubuntu Server from Windows

With the virtual machine running, the command `ssh username@IP-address` was run in Command Prompt on Windows. The fingerprint was accepted with yes, the password was entered, and the remote login was successful with the prompt `username@ubuntu`.

```
hina-ali32@ubuntu:~$ ssh hina-ali32@192.168.35.129
C:\Users\Hp>ssh hina-ali32@192.168.35.129
The authenticity of host '192.168.35.129 (192.168.35.129)' can't be established.
ED25519 key fingerprint is SHA256:4DoC3+ztPdQBe0vENMqD8zW8ERFwU/Q9i1brTixMJQY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Shows the SSH command.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Shows the fingerprint accepted.

```

C:\Users\Hp>ssh hina-ali32@192.168.35.129
The authenticity of host '192.168.35.129 (192.168.35.129)' can't be established.
ED25519 key fingerprint is SHA256:4DoC3+ztPdQBe0vENMqD8zW8ERFwU/Q9i1brTixMJQY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.35.129' (ED25519) to the list of known hosts.
hina-ali32@192.168.35.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Oct  5 05:57:52 PM UTC 2026

System load:  0.0           Processes:            219
Usage of /:   29.1% of 9.75GB Users logged in:          0
Memory usage: 11%          IPv4 address for ens33: 192.168.35.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

Shows the successful SSH login.

```

C:\Users\Hp>ssh hina-ali32@192.168.35.129
The authenticity of host '192.168.35.129 (192.168.35.129)' can't be established.
ED25519 key fingerprint is SHA256:4DoC3+ztPdQBe0vENMqD8zW8ERFwU/Q9i1brTixMJQY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.35.129' (ED25519) to the list of known hosts.
hina-ali32@192.168.35.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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System load:  0.0           Processes:            219
Usage of /:   29.1% of 9.75GB Users logged in:          0
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Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

hina-ali32@ubuntu:~$

```

Shows the remote prompt username@ubuntu.